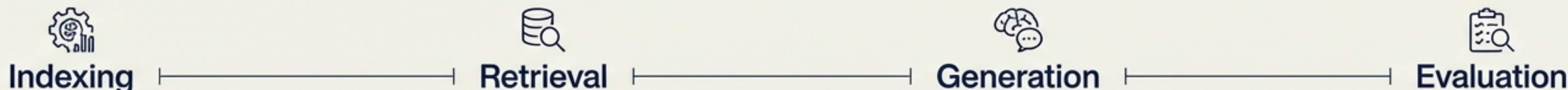
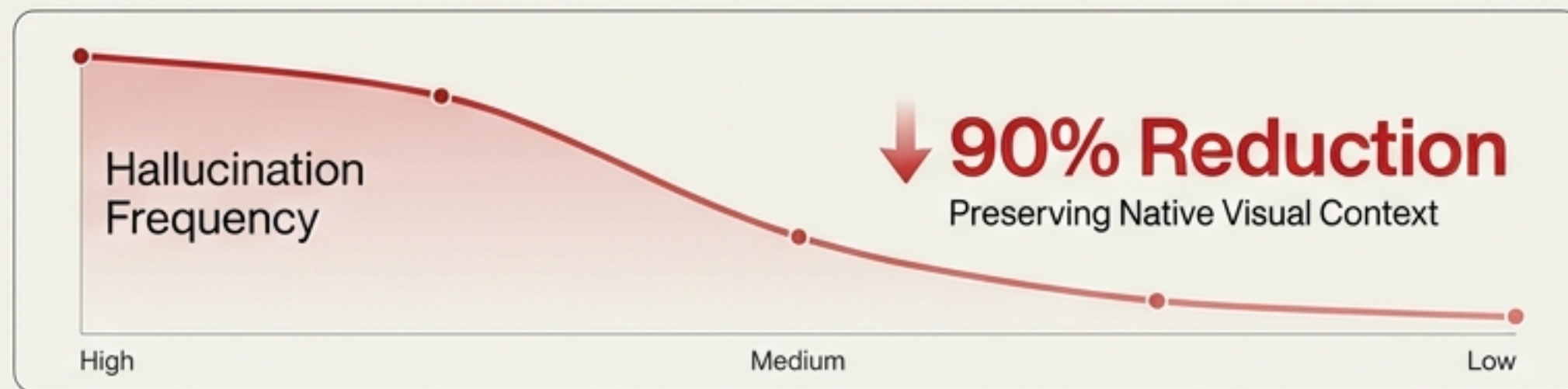
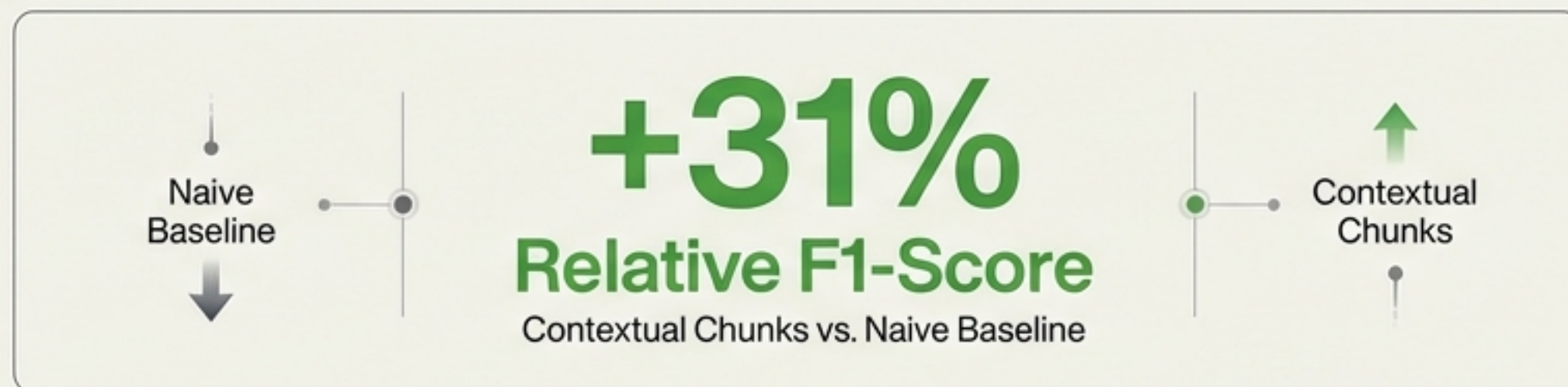
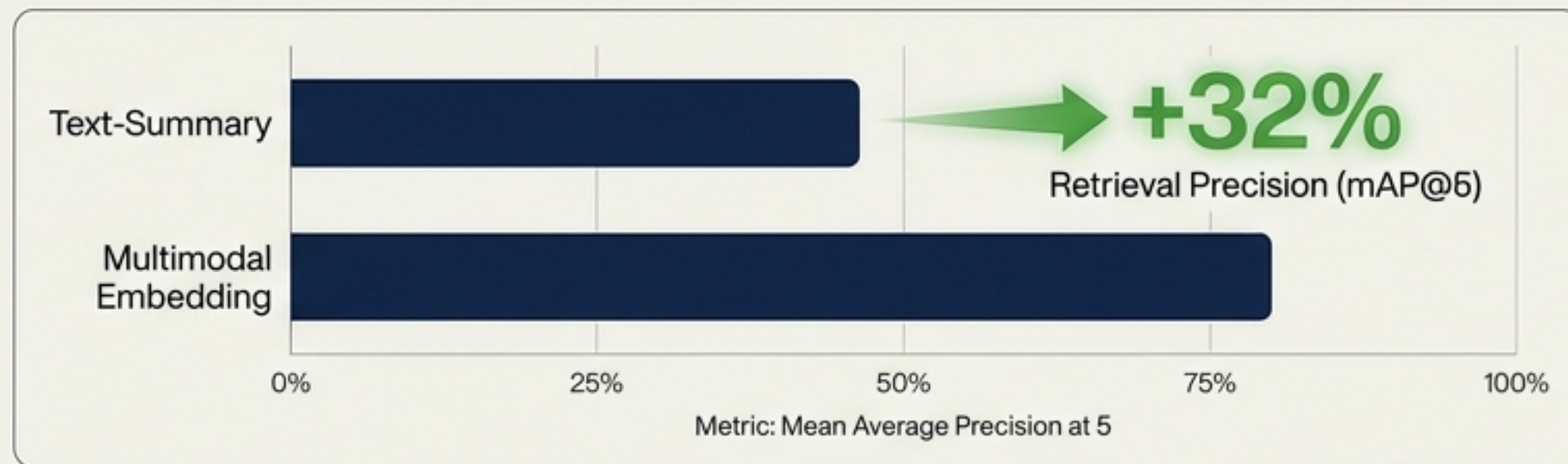


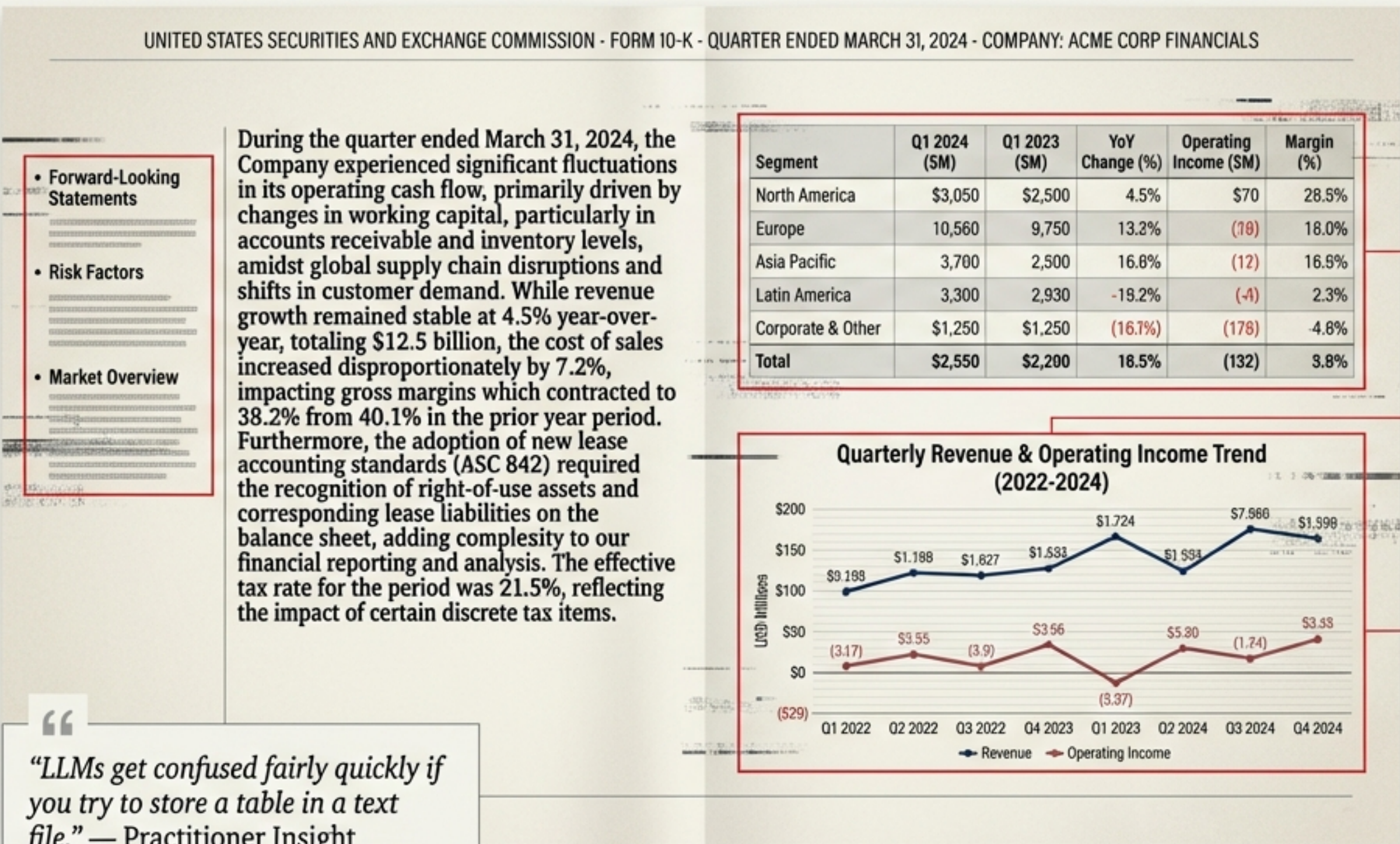
Beyond Naive RAG: A Blueprint for Financial & Complex Document Analysis

Synthesizing breakthroughs in Multimodal Embedding, Dynamic Context Selection, and Metadata-Driven Retrieval



The Financial Stress Test: Where Standard Pipelines Break

Analyzing the failures of automated parsing on complex financial documents.



- 1. The Table Problem: Flattening row/column relationships into text streams destroys semantic logic.**

Standard parsers fail to recognize the structural hierarchy, interpreting table cells as a sequence of unstructured text, losing crucial relational data.
- 2. The Layout Problem: Multi-column sidebars are parsed as “distractors”, scrambling the narrative flow.**

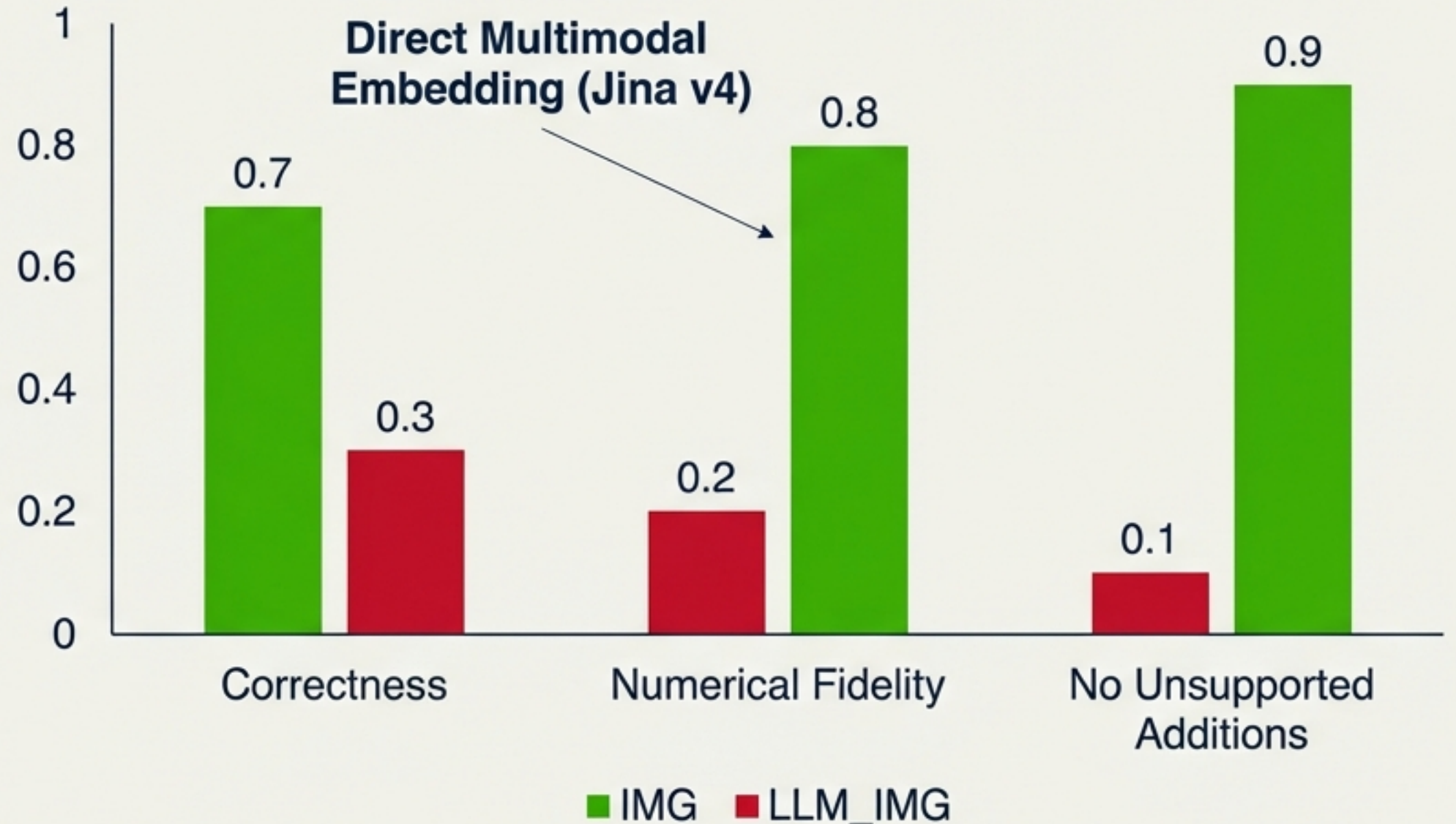
Parsers often treat sidebar content as main text, interspersing it randomly with the primary narrative, making sequential reading impossible.
- 3. The Conversion Loss: OCR/Captioning loses numerical precision before the LLM sees the data.**

Numerical values within charts and images are frequently misread by OCR or summarized vaguely by captioning models, discarding precise figures and trend data.

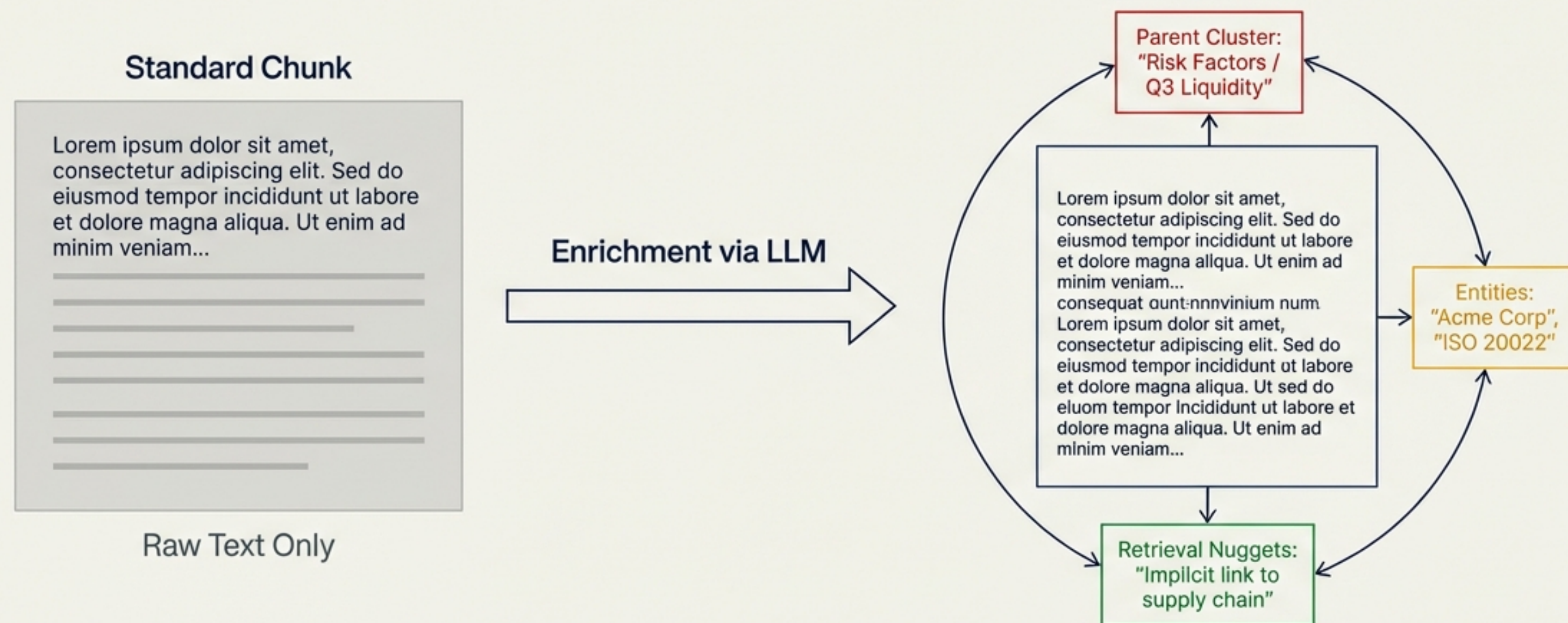
Stop Summarizing Charts. Start Embedding Them.

The Insight

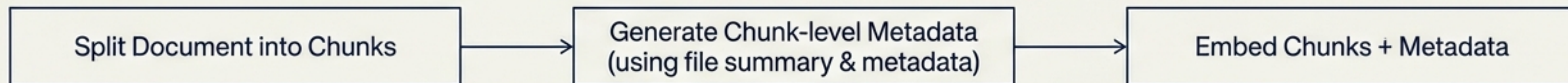
Standard Multimodal RAG relies on LLM captioning. This creates a bottleneck of information loss and fabricated details.



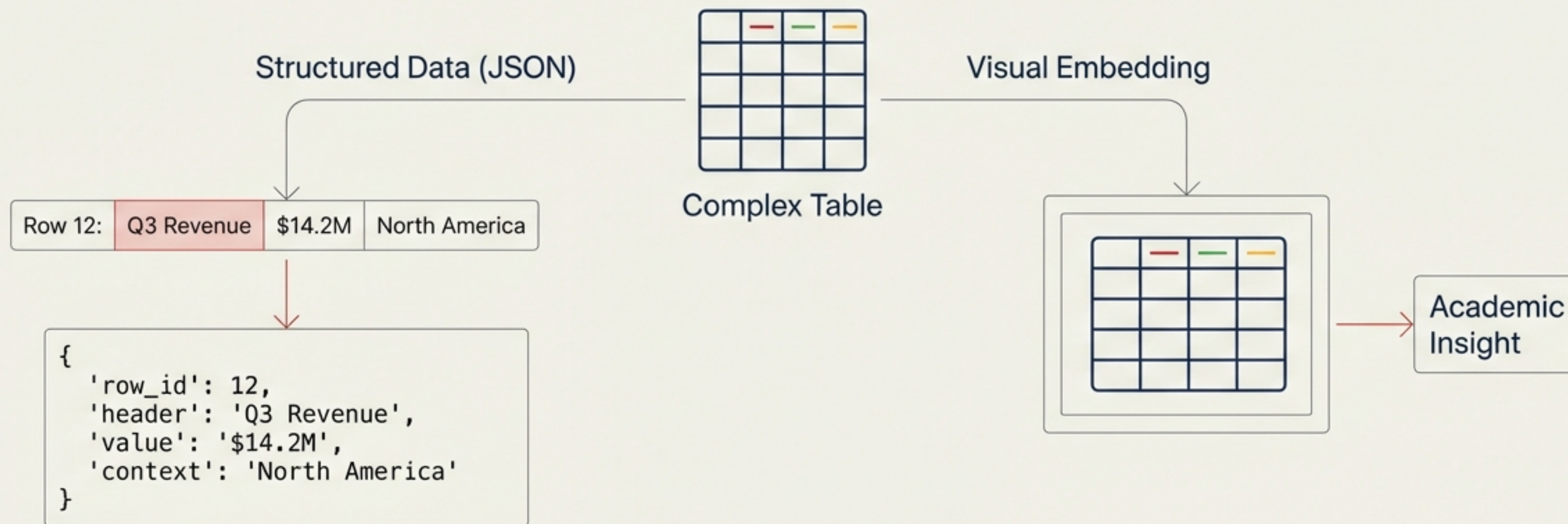
The Contextual Chunk: Baking Intelligence into the Index



Architectural Flow (Offline Indexing Pipeline Reference)



Handling Tables and Layouts.



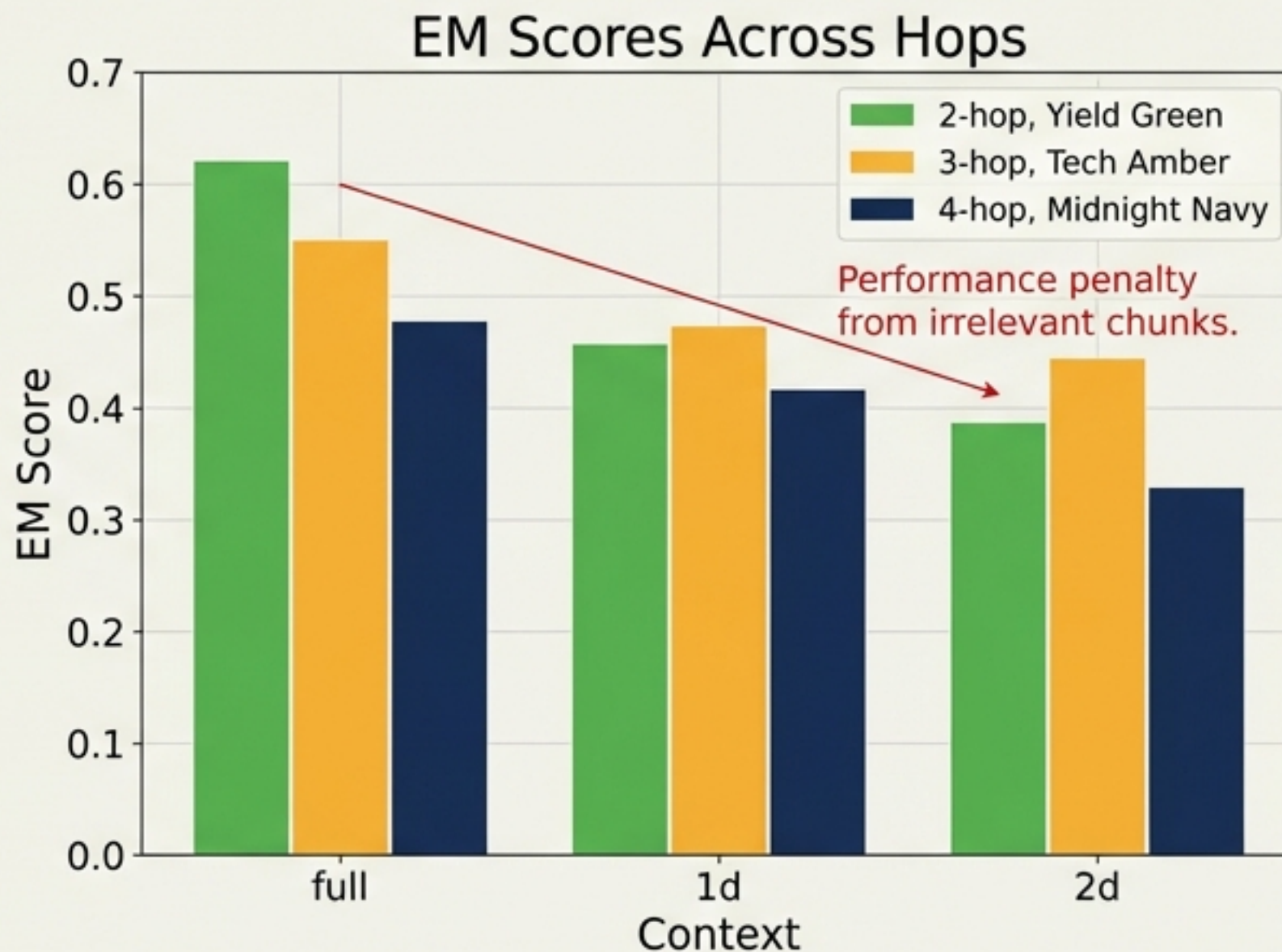
Practitioner Insight: Maintain key-value relationships by storing tables as structured JSON documents.

Academic Insight: For complex visual layouts, embed the table image directly (Multimodal Indexing).

Tooling Note: Use layout-aware parsers like Docling or Unstructured. Avoid simple text extraction.

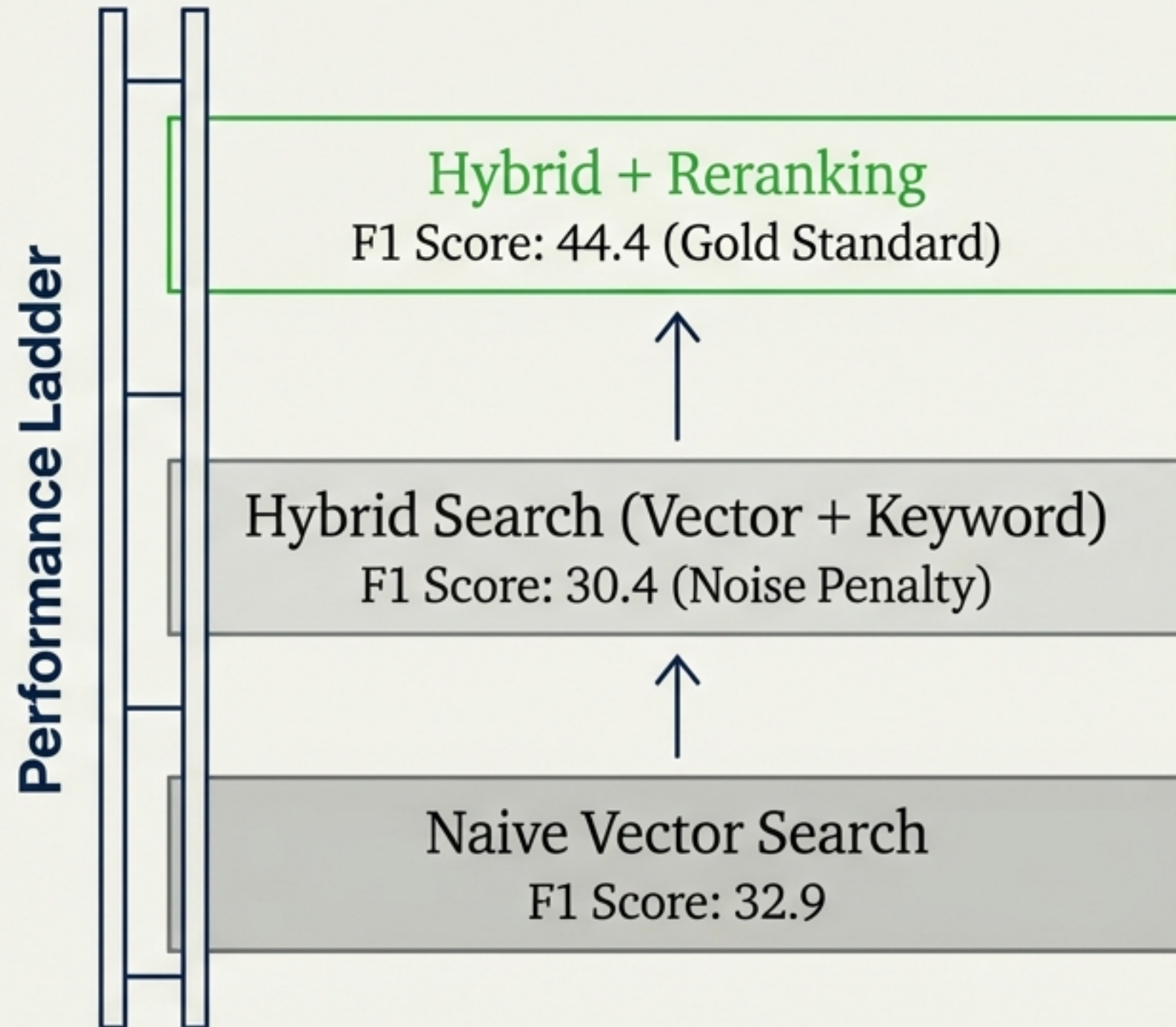
The 'Fixed-K' Fallacy: Why Static Retrieval Fails

- Simple queries need few chunks ($k=2$).
- Complex queries need broad context ($k=10$).
- Fixed-K (e.g., always $k=5$) introduces "Distractors".
- One distractor can drop generation accuracy by $>26\%$.



Solution: Dynamic Context Classifier. Predict optimal "k" per query.

Reranking is Non-Negotiable.



The Custom Metadata Reranker Recipe

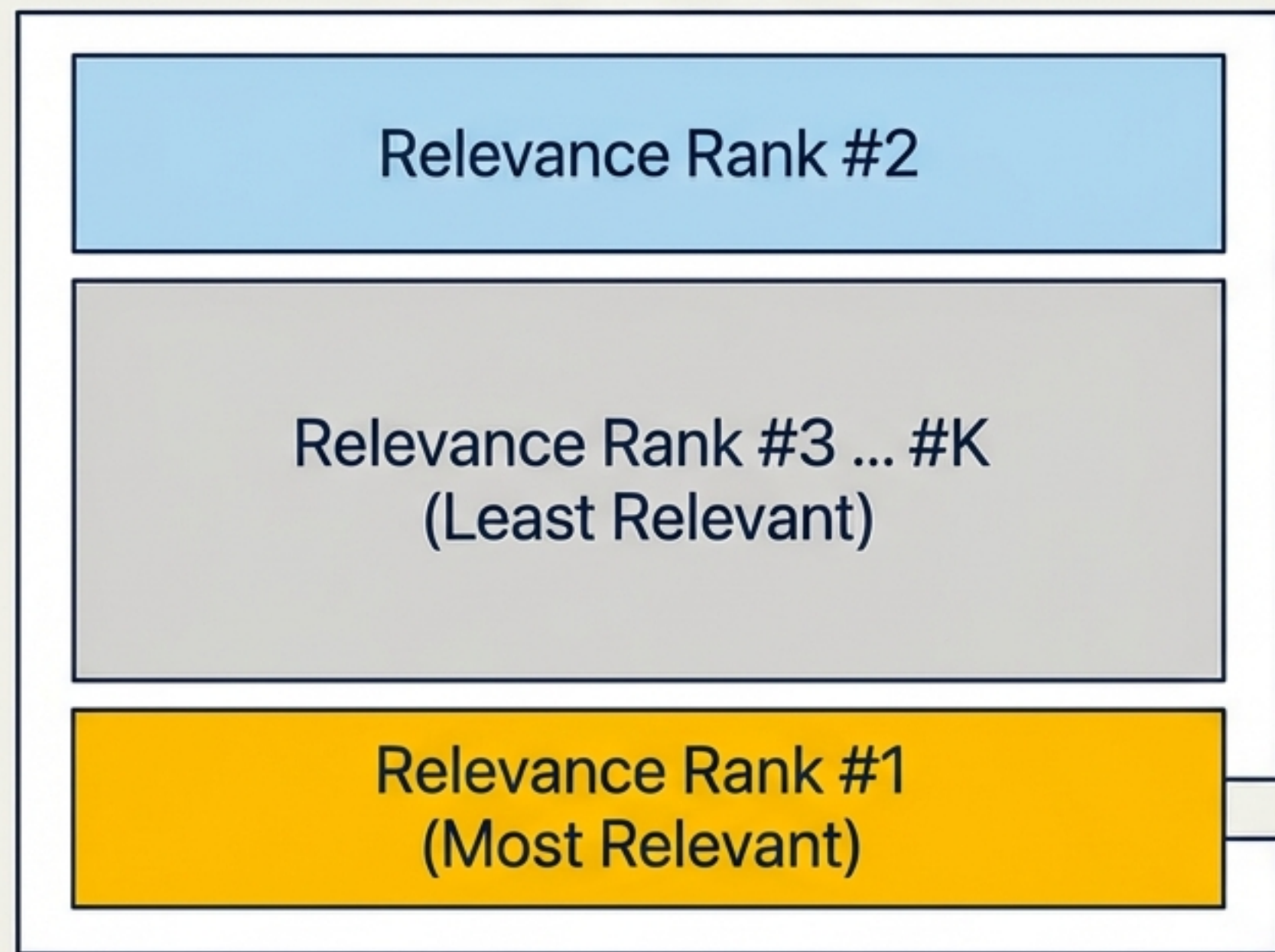
$$\begin{aligned}\text{Score} = & 0.25(\text{Vector Sim}) \\ & + 0.25(\text{Entity Match}) \\ & + 0.25(\text{Cluster Coherence}) \\ & + 0.25(\text{Query Alignment})\end{aligned}$$

Achieves F1 43.2 (vs 44.4 Commercial) at zero marginal cost.

Architecting the Prompt: Mitigating Positional Bias

Visualizing the Reverse-Relevant Ordering Strategy

LLM Context Window



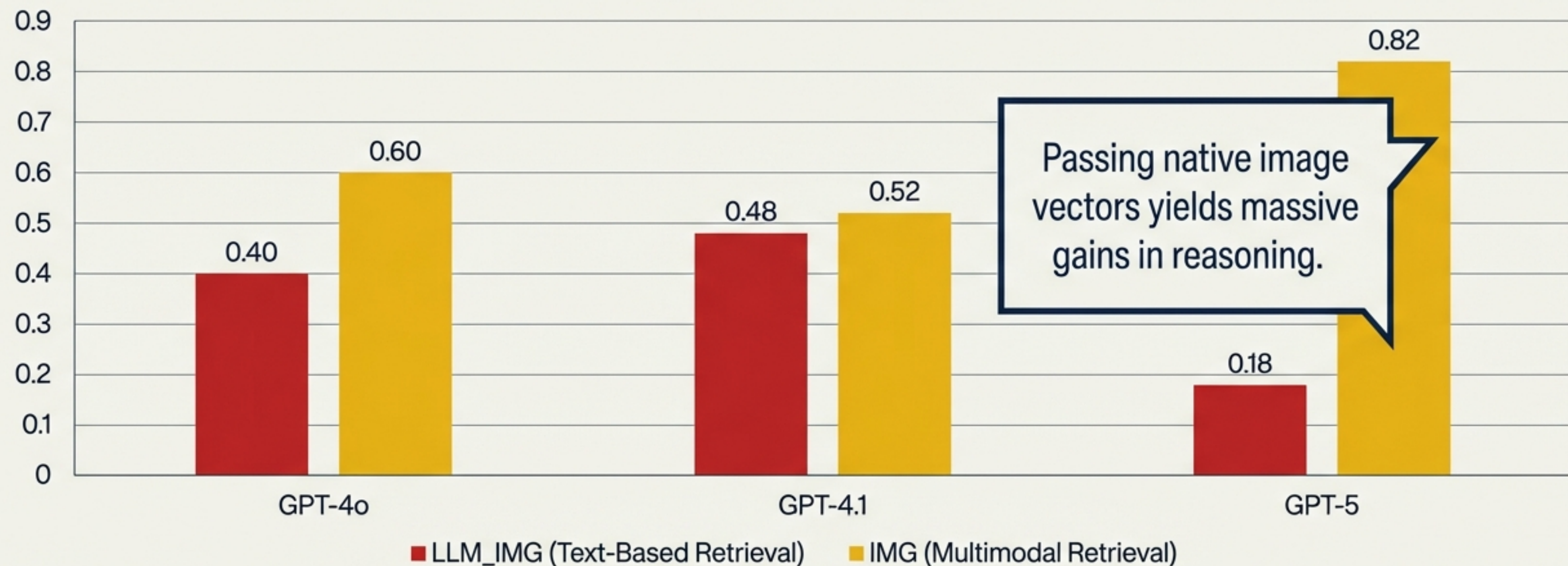
Exact Match Score:

0.5567 (End Position) vs
0.5391 (Middle Position).

Placing the most relevant context at the end of the prompt significantly improves attention and accuracy.

Highest Attention
Recency

Multimodal Generation: Fidelity through Native Inputs

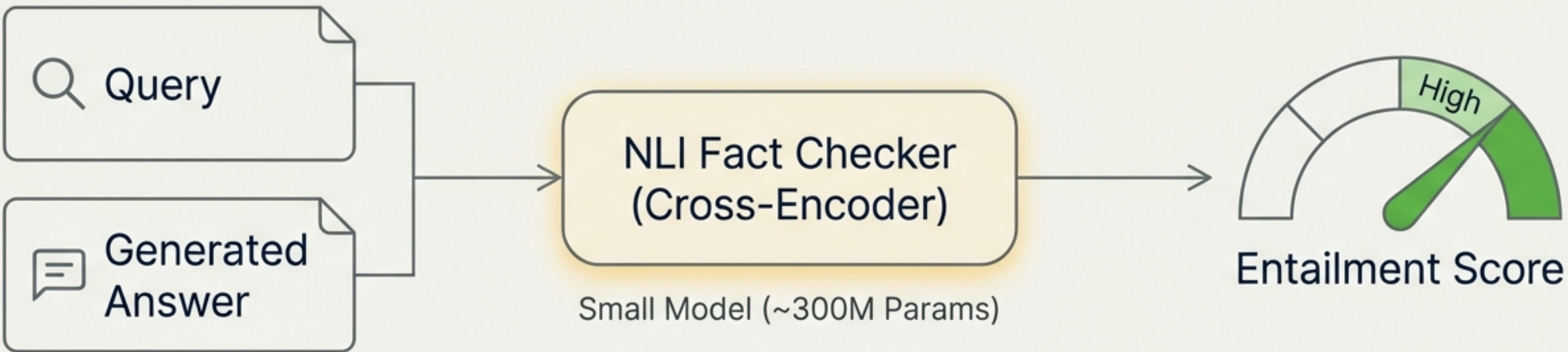


Clarity Score: 1.00 vs 0.00

Correctness: 0.70 vs 0.30

Trust but Verify: Lightweight Provenance Checking

Visualizing the NLI-Based Verification Process



Model: GPT-3.5-Turbo	AUC: 0.62	Latency: High
Model: Provenance (Ours)	AUC: 0.65+	Latency: Low



The Optimization Scorecard.

Indexing

Multimodal Embeddings

+32%

Relative improvement in Retrieval Precision (mAP@5)

Hallucinations

Visual Fidelity

-90%

Reduction in unsupported additions/hallucinations

Retrieval

Contextual Chunks + Rerank

+31%

Relative F1-Score improvement over naive baseline

Evaluation

NLI Verification

0.96 AUC

Fact verification performance using lightweight models

Implementation Anti-Patterns

Naive Chunk Expansion



Blindly adding neighboring chunks based on entity overlap adds noise.

Result: F1 dropped to 33.2;
Hallucinations spiked to 22%.

Text-Based Image Summarization



Using LLMs to caption charts destroys spatial/numerical precision.

Result: Loss of 0.60 points in
Numerical Fidelity.

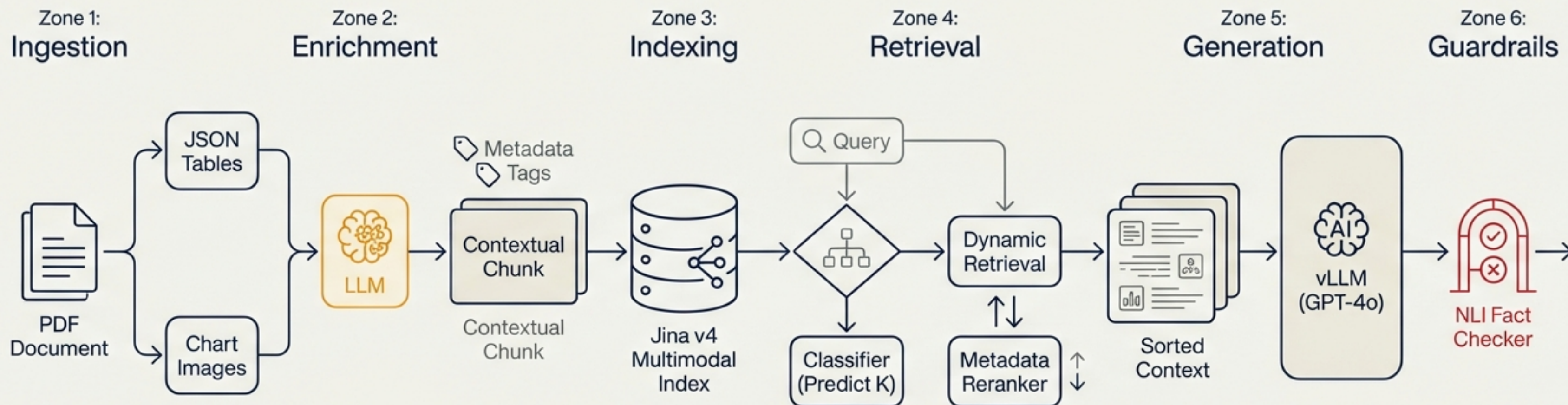
Hybrid Search w/o Reranking



Keyword search alone introduces irrelevant distractors.

Result: Performance worse than
dense-only baseline (F1 30.4).

The State-of-the-Art Architecture.



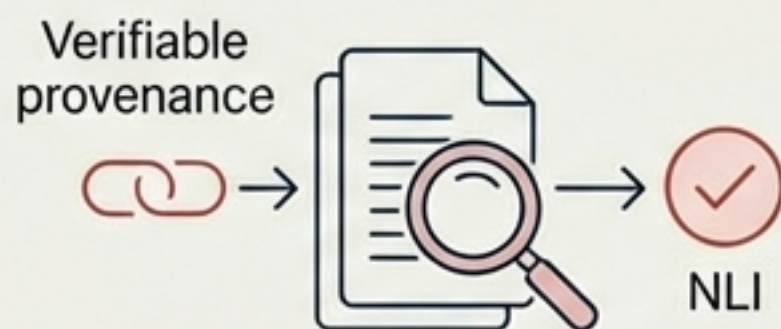
From Naive to Native.

Accuracy



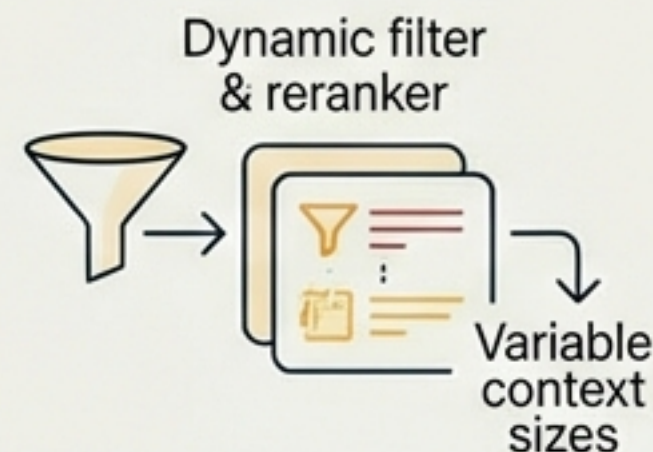
30%+ gains by preserving native structure (Tables, Images) and using metadata-driven reranking.

Trust



Verifiable provenance and hallucination control through NLI checks, not just LLM vibes.

Efficiency



Cost-effective custom rerankers and dynamic context sizing to reduce token waste.

Preserving information in its native form—whether visual, tabular, or structural—is the only path to reliable complex reasoning.

References.

1. Lumer et al. (PwC) - “Comparison of Text-Based and Image-Based Retrieval in Multimodal RAG Systems”
2. Iratni et al. (IRIT) - “Dynamic Context Selection for Retrieval-Augmented Generation”
3. Dadopoulos et al. - “Metadata-Driven Retrieval-Augmented Generation for Financial Question Answering”
4. Sankararaman et al. (Uniphore) - “Provenance: A Light-weight Fact-checker for Retrieval Augmented LLM Generation”
5. Practitioner Discussion: r/LangChain Community Threads on PDF/Table Parsing